Service kits - design brief

AI BRAKING / AB-SP / C0 / 2026-09-07

An identifiable caliper service set: backing plates, lining blocks, retainer pins and bushes. It is scoped to AI Braking concepts; no fit or compatibility with a competitor brake is implied.

TARGETS (not validated)

Pad footprint: 70 × 60 mm. Shared with AB-DC

Lining concept thickness: 8 mm. Wear allowance unapproved

Retainer pin diameter: 8 mm. Nominal concept

Traceability: 100% kits linked to parent revision. Process objective, not achieved factory evidence

DESIGN DECISIONS

Separate wear parts from structural parts in the BOM and drawing numbering.

Source the friction compound and its retention process from a specialist.

Mark non-friction backing surfaces, avoiding stress-sensitive and rubbing surfaces.

CANDIDATE MATERIALS

Backing plate: C45 candidate, corrosion protection excluded from bond/friction areas.

Lining: supplier-controlled non-asbestos compound, not a generic material grade.

Pins/bushes: certified steel and purchased bronze candidates.

Assembly and engineering gates

ASSEMBLY INTENT

1. Verify parent brake revision and compound lot.
2. Inspect backing plate and clean bond surfaces using supplier process.
3. Bond/retain lining only to an approved supplier procedure.
4. Install pins and bushes with the engineer-approved fit and retention.

OPEN DESIGN RISKS

1. Friction coefficient changes with pressure, speed, temperature and wear.
2. Bond strength, rivet design, retainer fatigue and wear limits are not specified.

VALIDATION REQUIRED

1. Bond/retention strength, dimensional fit and compound lot consistency.
2. Paired dynamometer testing of fade, recovery and wear.

ELECTRICAL / SENSOR REQUIREMENTS ONLY

1. Record digital part/lot identity and wear measurement requirements; sensor design deferred.

Assembly reference / numbered components

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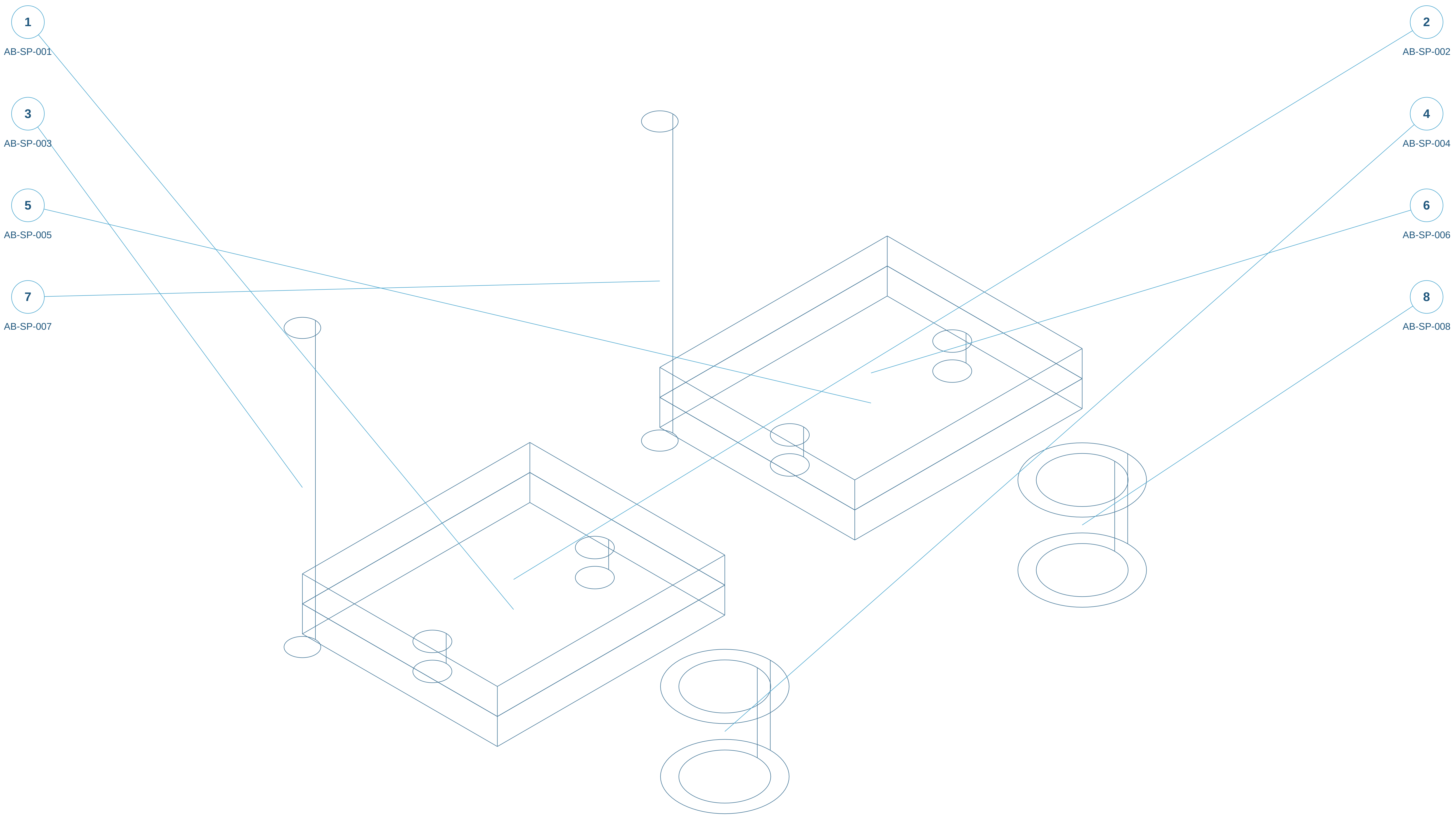

Static isometric reference with item marks. Exploded geometry and part selection are available in the web workspace.

ENGINEER REVIEW ONLY - NOT FOR MANUFACTURE - NO VALIDATED RATINGS

Original concept geometry. Nominal mm. No tolerances, GD&T, finish, preload, material approval or certification.

Parts and interfaces

AB-SP-001 | Backing plate 1 | make-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-SP-002 | Lining block 1 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Specialist friction supplier

AB-SP-003 | Retainer pin 1 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Turn/grind; retention detail pending

AB-SP-004 | Bush 1 | make-concept

Purchased bronze bearing candidate; Buy or turn to approved fit

AB-SP-005 | Backing plate 2 | make-concept

C45 candidate; Saw/laser blank; CNC critical faces and bores

AB-SP-006 | Lining block 2 | bought-envelope

Supplier-controlled; illustrative reserved envelope; Specialist friction supplier

AB-SP-007 | Retainer pin 2 | make-concept

42CrMo4 candidate; heat treatment / fit unapproved; Turn/grind; retention detail pending

AB-SP-008 | Bush 2 | make-concept

Purchased bronze bearing candidate; Buy or turn to approved fit

AB-SP-001 / Backing plate 1

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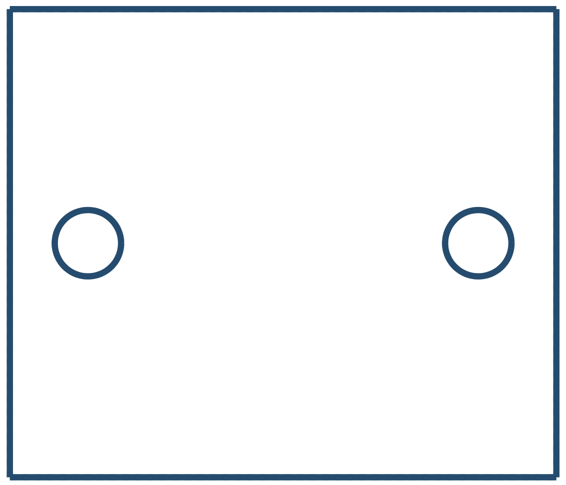
TOP X-Y



C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

70 x 60 x 8; 2 x diameter 8.5 on 50 mm pitch.

Common nominal planform with AB-DC.

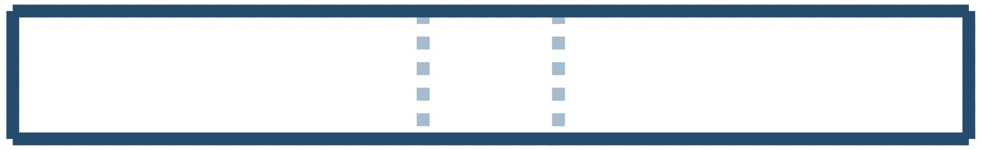
CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z



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AB-SP-002 / Lining block 1

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Supplier-controlled; illustrative reserved envelope

Specialist friction supplier

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

8 mm concept thickness; compound and retention unresolved.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z

SIDE Y-Z

AB-SP-003 / Retainer pin 1

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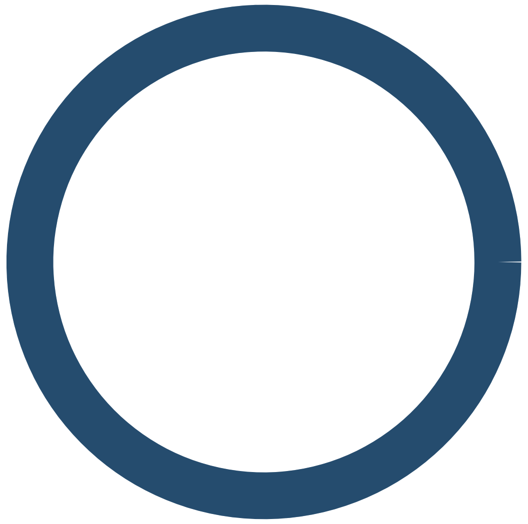
TOP X-Y



42CrMo4 candidate; heat treatment / fit unapproved

Turn/grind; retention detail pending

Overall X/Y/Z bounds: 8.00 / 8.00 / 85.00 mm

Diameter 8 x 85 smooth envelope; no final retaining feature.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

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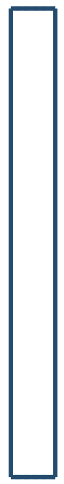
Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z

SIDE Y-Z



AB-SP-004 / Bush 1

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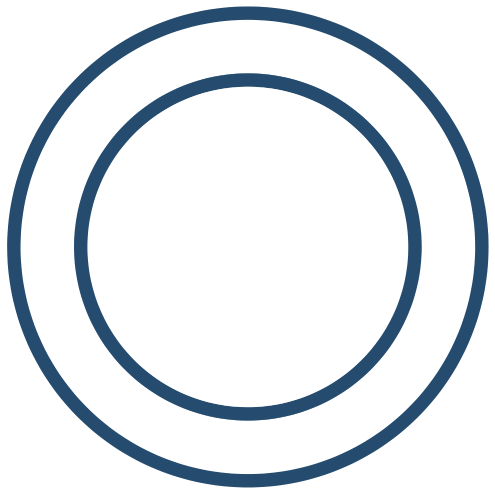
TOP X-Y



Purchased bronze bearing candidate

Buy or turn to approved fit

Overall X/Y/Z bounds: 28.00 / 28.00 / 24.00 mm

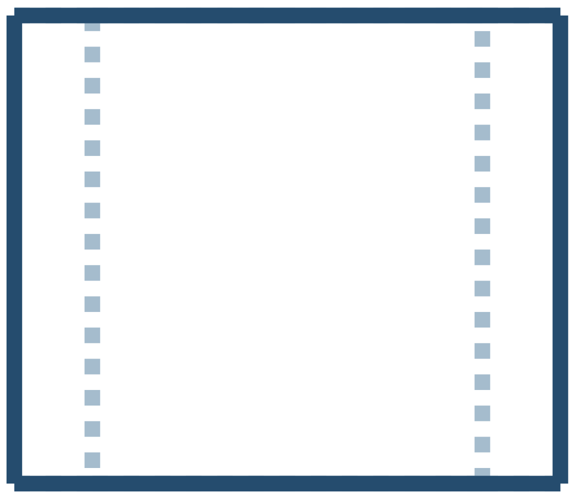
OD 28; bore 20; length 24; fit not assigned.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

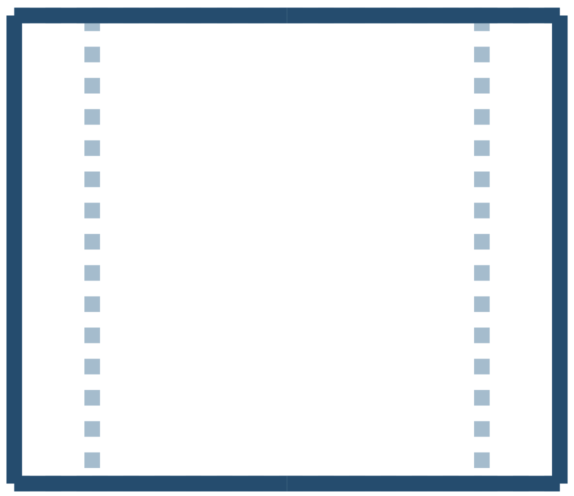
Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

FRONT X-Z



SIDE Y-Z



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AB-SP-005 / Backing plate 2

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C45 candidate

Saw/laser blank; CNC critical faces and bores

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

70 x 60 x 8; 2 x diameter 8.5 on 50 mm pitch.

Common nominal planform with AB-DC.

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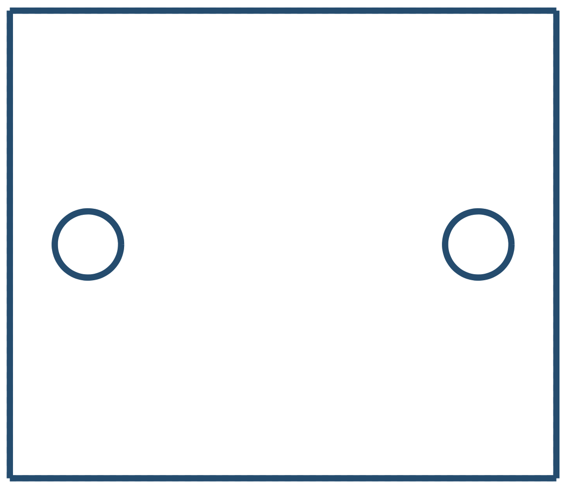
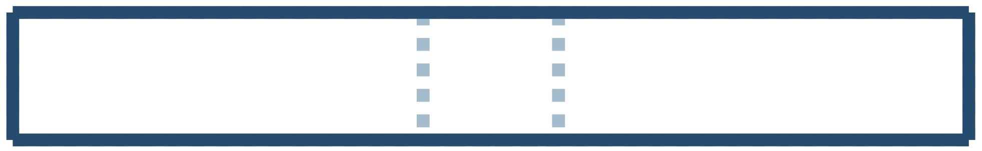
Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y



FRONT X-Z

SIDE Y-Z



AB-SP-006 / Lining block 2

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Supplier-controlled; illustrative reserved envelope

Specialist friction supplier

Overall X/Y/Z bounds: 70.00 / 60.00 / 8.00 mm

8 mm concept thickness; compound and retention unresolved.

CAD hidden-line reference projections. Bounds and selected features are nominal; see STEP for solid geometry.

Datum proposal: choose primary mounting face A, functional bore/axis B and clocking feature C during review.

Part marking: item ID + C0 + material lot on a non-friction/non-fit surface; exact location unresolved.

TOP X-Y

FRONT X-Z

SIDE Y-Z

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AB-SP-007 / Retainer pin 2

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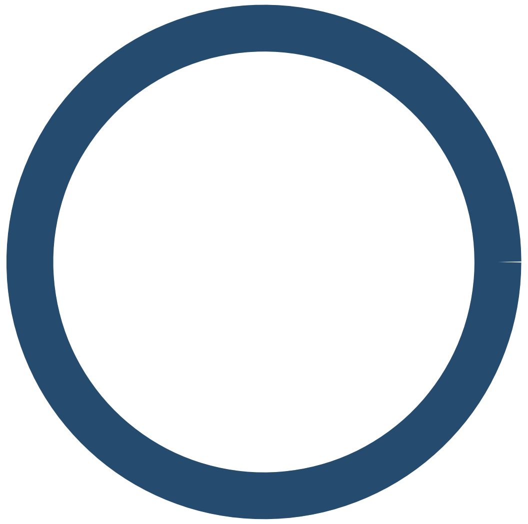
TOP X-Y



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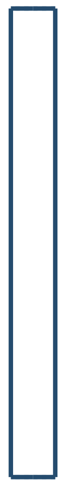
FRONT X-Z

SIDE Y-Z



AB-SP-008 / Bush 2

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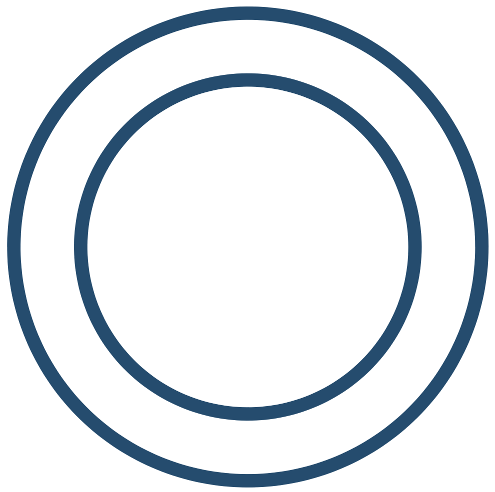
TOP X-Y



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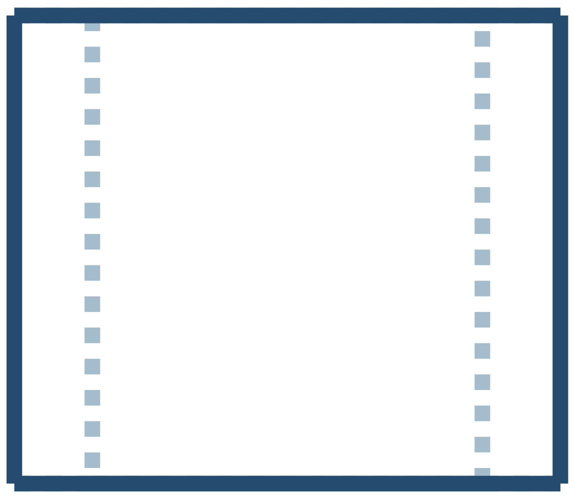
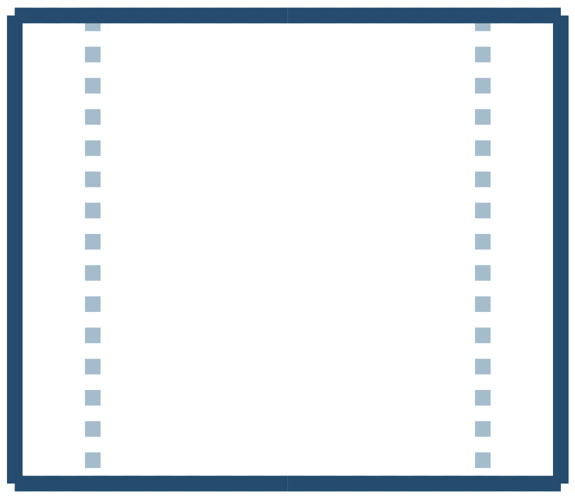
FRONT X-Z



SIDE Y-Z



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